

Introduction to Python and Data Structures

Python is one of the most popular programming languages in the world. It is widely used in fields such as web development, data analysis, artificial intelligence, and automation. One of Python's strengths is its rich set of built-in data structures that make programming more efficient and intuitive.

Common Data Structures in Python

1. List: An ordered, mutable collection of elements.
2. Tuple: An ordered, immutable collection of elements.
3. Dictionary: A collection of key-value pairs, allowing fast lookups.
4. Set: An unordered collection of unique elements.

Example Code

```
numbers = [1, 2, 3, 4]
numbers.append(5)
print('List:', numbers)

person = {'name': 'Alice', 'age': 25}
print('Dictionary:', person)

unique_items = set([1, 2, 2, 3, 3, 4])
print('Set:', unique_items)
```

Conclusion

Understanding Python's core data structures is essential for writing clean, efficient, and effective code. Mastery of these structures provides a strong foundation for solving complex problems and developing robust applications.